Python FastAPI Coding Exercise

Introduction

BEON High School is building a system to manage users, courses and enrollments.
Currently, the system only has a basic user module implemented.

Your task is to extend the system with a courses module and enrollment functionality, while properly enforcing business rules and handling errors.

This exercise is designed to test your ability to:

- Implement REST APIs in Python (FastAPI recommended)
- Work with relational databases
- Enforce business rules
- Handle errors and edge cases
- Write clean and maintainable code

Existing Module

User module: The user model has the following fields:

- first_name
- last_name

You can use this module as a reference.

Exercise Tasks

1. **Courses module:** Implement full CRUD operations for courses:
    - Create course
    - List courses
    - Get course by ID
    - Edit course
    - Delete course

Course model fields:

- title
- description

2. Enrollment module: Implement the ability for users to enroll in courses

- A user can be enrolled in up to 5 courses.
- A course can have up to 20 users.

3. Error handling: Anticipate and manage invalid operations, missing resources or other exceptional cases in a way that makes the API reliable and user-friendly.

4. Authentication and authorization

- Extend the user model with email, password and role.
- Users should be able to log in and perform actions according to their role.
- Consider how different roles might have different access levels for reading, creating, editing and deleting.

Additional Notes

- You can use SQLAlchemy and any relational database (like PostgreSQL).
- Ensure the relationships between users, courses and enrollments are properly normalized.
- Unit tests are encouraged to validate business rules and error handling.